Task 1: API Integration & Data Visualization

```
import requests
import matplotlib.pyplot as plt

API_KEY = "your_api_key_here"
city = "Chennai"
url =
f"http://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}&units=metr
ic"

response = requests.get(url)
data = response.json()

temp = data['main']['temp']

plt.bar(city, temp, color='skyblue')
plt.ylabel("Temperature (°C)")
plt.title("Current Temperature")
plt.show()
```

◈ Task 2: Automated Report Generation

```
import pandas as pd
from fpdf import FPDF

df = pd.read_csv("data.csv")
summary = df.describe()

pdf = FPDF()
pdf.add_page()
pdf.set_font("Arial", size=12)

pdf.cell(200, 10, txt="Automated Report", ln=True, align="C")
pdf.multi_cell(0, 10, txt=str(summary))

pdf.output("report.pdf")
```

◈ Task 3: AI Chatbot with NLP

```
import nltk
from nltk.chat.util import Chat, reflections

pairs = [
    [r"hi|hello", ["Hello! How can I help you today?"]],
    [r"what is your name?", ["I am your internship chatbot."]],
    [r"bye", ["Goodbye! Have a great day."]]
]

chatbot = Chat(pairs, reflections)
chatbot.converse()
```

◈ Task 4: Machine Learning Model Implementation

```
from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import CountVectorizer

from sklearn.naive_bayes import MultinomialNB

from sklearn.metrics import accuracy_score


texts = ["Win money now!", "Hello friend, how are you?", "Claim your prize!!!", "Meeting at 5pm"]

labels = [1, 0, 1, 0]  # 1 = spam, 0 = not spam


vectorizer = CountVectorizer()

X = vectorizer.fit_transform(texts)


X_train, X_test, y_train, y_test = train_test_split(X, labels, test_size=0.5, random_state=42)


model = MultinomialNB()

model.fit(X_train, y_train)


y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
```